



CRANBROOK
SCHOOL

Centre Number

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Student Number

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2021

HSC Examination

Assessment Task 4

Extension 2 Mathematics

Trial Examination

General Instructions

- Write using blue or black pen
- Diagrams drawn using pencil
- A Board-approved calculator may be used
- **Section 1:** Use the **Multiple Choice** Answer sheet for questions 1 to 10.
- **Section 2:** Please write each question in a new booklet.
- All relevant working should be shown for each question.

This paper must not be removed from the examination room

Disclaimer

The content and format of this paper does not necessarily reflect the content and format of the HSC examination paper.

Section I – Multiple Choice

10 Marks

1. Which of the following points lies on the line described by the vector equation

$$r = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} ?$$

(A) $\begin{pmatrix} -3 \\ 9 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} -3 \\ -8 \\ -3 \end{pmatrix}$

(C) $\begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$

(D) $\begin{pmatrix} 3 \\ -5 \\ 2 \end{pmatrix}$

2. Which value of z satisfy $z^2 = 7 + 24i$?

(A) $4 + 3i$

(B) $-4 + 3i$

(C) $-3 + 4i$

(D) $3 + 4i$

3. What is the angle between vectors $\underline{u} = 2\underline{i} - \underline{j} + \underline{k}$ and $\underline{v} = \underline{i} + 3\underline{j} + 2\underline{k}$, to the nearest degree?

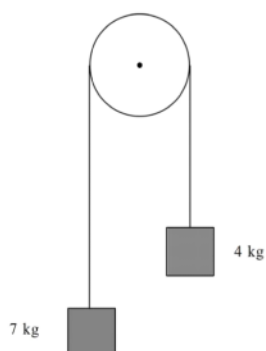
- (A) 77°
- (B) 83°
- (C) 84°
- (D) 96°

4. $A(1, 2, 2)$, $B(3, -12, 4)$, $C(1, 2, 0)$ and $D(3, -12, 0)$ are four positional vectors.

What is the vector projection of $\overline{AB}$ onto $\overline{CD}$?

- (A) $2\underline{i} - 14\underline{j} + 2\underline{k}$
- (B) $2\underline{i} - 14\underline{j} + 4\underline{k}$
- (C) $2\underline{i} - 14\underline{j}$
- (D) $-2\underline{i} + 14\underline{j}$

5. A light inextensible string passes over a smooth pulley. Attached to each end of the strings are masses of 4 kg and 7 kg, as shown.



The acceleration of the larger mass downwards is

- (A) $\frac{3g}{11}$
- (B) $\frac{11g}{3}$
- (C) $\frac{7g}{11}$
- (D) $3g$

6. A particle is moving in simple harmonic motion with a displacement of x metres. Its acceleration, $\ddot{x}$, is given by $\ddot{x} = -4x + 3$.

What are the centre and period of motion?

- (A) Centre of motion = 3, period = $\frac{\pi}{2}$
- (B) Centre of motion = -3, period = π
- (C) Centre of motion = $\frac{3}{4}$, period = π
- (D) Centre of motion = $\frac{3}{4}$, period = $\frac{\pi}{2}$

7. It is given that $z = 2 + i$ is a root of $z^3 + az^2 - bz + 5 = 0$, where a and b are real numbers.

What is the value of a ?

- (A) -5
- (B) -3
- (C) 3
- (D) 5

8. Which integral has the smallest value?

(A) $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$

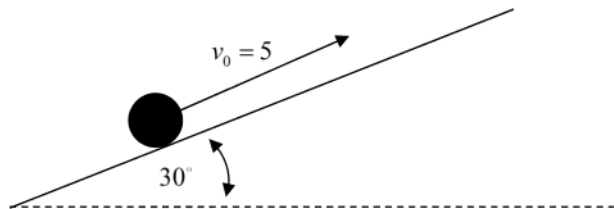
(B) $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$

(C) $\int_0^{\frac{\pi}{4}} \sin x \cos x \, dx$

(D) $\int_0^{\frac{\pi}{4}} \sin x \tan x \, dx$

9. A ball of unit mass is rolled up a frictionless ramp that is inclined at 30° to the horizontal. It has an initial velocity of 5 ms^{-1} .

Assuming $g = 10 \text{ ms}^{-2}$, what is the net acceleration on the ball?



- (A) $5\sqrt{3} \text{ ms}^{-2}$ directed down the ramp.
(B) $5\sqrt{3} \text{ ms}^{-2}$ directed up the ramp.
(C) 5 ms^{-2} directed down the ramp.
(D) 5 ms^{-2} directed up the ramp.

10. What value of a will minimise the integral $\int_0^1 (x^2 - a)^2 dx$?

(A) $a = \frac{1}{2}$

(B) $a = \frac{1}{\sqrt{2}}$

(C) $a = \frac{4}{45}$

(D) $a = \frac{1}{3}$

END OF MULTIPLE CHOICE

Section II – Extended response

90 Marks

Question 11 – Please start a new booklet.

15 Marks

(a) Let $z = 4 - 3i$ and $w = 2 + 5i$, evaluate

(i) Show that $\frac{w}{\bar{z}} = \frac{23 + 14i}{25}$. 2

(ii) Show that $(w + \bar{z})(\bar{w} + z) = 100$. 2

(b) Find the square roots of $15 - 8i$. Show all working. 3

(c) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta + \cos \theta}$. 4

- (d) The acceleration, a , of a particle moving in a straight line is given by $a = x - 4$ where x is the displacement in metres. The particle is initially at the origin and travelling with velocity of 2 ms^{-1} .
- (i) Show that the velocity of the particle is described by $v^2 = x^2 - 8x + 4$. **2**
- (ii) Find the acceleration of the particle when it comes to rest. **2**

End of Question 11

Question 12 - Please start a new booklet.

15 Marks

(a) Use integration by parts to find $\int x3^x dx$ **3**

(b) By writing $\frac{8-2x}{(1+x)(4+x^2)}$ in the form $\frac{a}{1+x} + \frac{bx+c}{4+x^2}$, evaluate **4**

$$\int_0^4 \frac{8-2x}{(1+x)(4+x^2)} dx$$

(c)

(i) On the same Argand diagram, draw a neat sketch of $|z-4-4i|=2$ and

$$\arg(z) = \frac{\pi}{4}. \quad \text{2}$$

(ii) Hence write down all the values of z which satisfy $|z-4-4i|=2$ and $\arg(z) = \frac{\pi}{4}$

simultaneously. **2**

(d) Find the scalar projection of the vector $\underline{u} = \underline{i} - 2\underline{j} + \underline{k}$ onto the vector $4\underline{i} - 4\underline{j} + 7\underline{k}$. **2**

(e) Given $\underline{a} = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix}$, and $\underline{a} - \underline{b} + 2\underline{c} = 0$, find $\underline{c}$. **2**

End of Question 12

Question 13 – Please start a new booklet.**15 Marks**

(a) Let $I_1 = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$ and $I_2 = \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$.

(i) Using the substitution $u = \pi - x$, show that $I_1 = I_2$ **2**

(ii) Hence, or otherwise, evaluate I_1 . **2**

(b) A mass of 1 kg moves along a straight line with velocity $v \text{ m s}^{-1}$. It encounters a resistance of $v + v^3$. The particle has initial velocity U , where $U > 0$ and starts from the origin.

(i) Show that the equation of motion is $\ddot{x} = -v(1 + v^2)$. **1**

(ii) Show that $x = \tan^{-1}\left(\frac{U - v}{1 + Uv}\right)$. **2**

(iii) Show that $v^2 = \frac{U^2}{(1 + U^2)e^{2t} - U^2}$. **3**

- (c) At time t the particle has velocity v and displacement x . A particle is travelling in a straight line. Its displacement, x cm, from O at a given time, t seconds after the start of the motion, is given by $x = 3 + \sin^2 t$.
- (i) Prove that the particle is undergoing simple harmonic motion. **2**
- (ii) Find the period of the motion. **1**
- (iii) Find the total distance travelled by the particle in the first π seconds. **2**

End of Question 13

Question 14 – Please start a new booklet.

15 Marks

- (a) The scalar product of $\underline{i} - 2\lambda\underline{j} - \underline{k}$ and the sum of $\underline{i} - \lambda\underline{k}$ and $\lambda\underline{i} + 2\underline{j} - \underline{k}$, is 6.

Find λ .

2

- (b) Relative to the origin O , the points A , B , C and D have position vectors given respectively by $-4\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$, $4\mathbf{i} + \lambda\mathbf{j} + 6\mathbf{k}$, $4\mathbf{i} - \mathbf{j} - \mathbf{k}$ and $2\mathbf{j} - 6\mathbf{k}$.

- (i) Given that the line $\overline{AC}$ is perpendicular to the line $\overline{BD}$, determine the value of λ .

2

- (ii) Hence find the position vector of F , the point of intersection of the lines $\overline{AC}$ and $\overline{BD}$.

3

- (c) Let $\overline{OA} = \underline{a}$, $\overline{OB} = \underline{b}$ and $\overline{OC} = 3\underline{a} + 2\underline{b}$.

- (i) Prove that if $\overline{OD} = \frac{1}{5}\overline{OC}$, the D lies on AB .

3

- (ii) Is the point D closer to point A or point B ? Justify your answer.

1

(d)

(i) Given $z = \cos \theta + i \sin \theta$, prove that $z^n - \frac{1}{z^n} = 2i \sin n\theta$. **1**

(ii) Hence, by considering the expansion of $\left(z - \frac{1}{z}\right)^5$, show that **3**

$$\sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$$

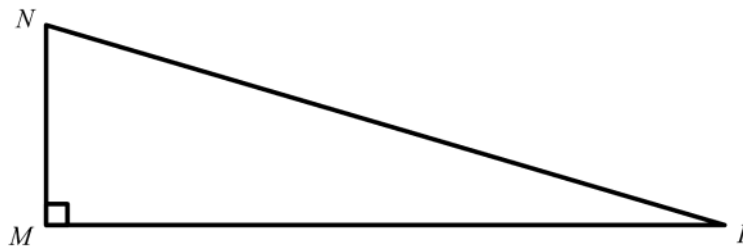
End of Question 14

Question 15 – Please start a new booklet.

15 Marks

(a)

- (i) Prove that for non-zero vectors, $\underline{a}$ and $\underline{b}$, $(\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) = |\underline{a}|^2 + |\underline{b}|^2$ if $\underline{a}$ and $\underline{b}$ are perpendicular. 2
- (ii) In $\triangle LMN$, let $\overrightarrow{LM} = \underline{a}$ and $\overrightarrow{MN} = \underline{b}$.



By finding an expression for the side LN in terms of vectors $\underline{a}$ and $\underline{b}$, or otherwise, prove that $|LN|^2 = |LM|^2 + |MN|^2$. 2

- (b) Find $\int x^2 \sqrt{1-x^2} \, dx$ 3

- (c) A particle of unit mass is moving vertically downward in a medium which exerts a resistance force proportional to the square of the speed, v , of the particle. It is released from rest at O and its terminal velocity is U .

- (i) Show that the distance it has fallen below O is given by **2**

$$x = \frac{1}{2k} \ln \left| \frac{g}{g - kv^2} \right|.$$

- (ii) Prove that the time taken, T , for the particle to fall from O to when its velocity is half its terminal velocity, U , is given by **3**

$$T = \frac{U}{2g} \ln 3.$$

- (d) Using integration by parts, calculate $\int (1 + 2x^2) e^{x^2} dx$. You may wish to consider this integral as the sum of two integrals. **3**

End of Question 15

Question 16 – Please start a new booklet.**15 Marks**

(a)

- (i) Find all the roots of $z^7 - 1 = 0$ in exponential form. 2

- (ii) Using $z^7 - 1 = (z - 1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$, or otherwise,

prove that $\frac{2\pi}{7}$, $\frac{4\pi}{7}$ and $\frac{6\pi}{7}$ are solutions to 3

$$2 \cos 3\theta + 2 \cos 2\theta + 2 \cos \theta + 1 = 0$$

- (iii) Hence, or otherwise, prove that 2

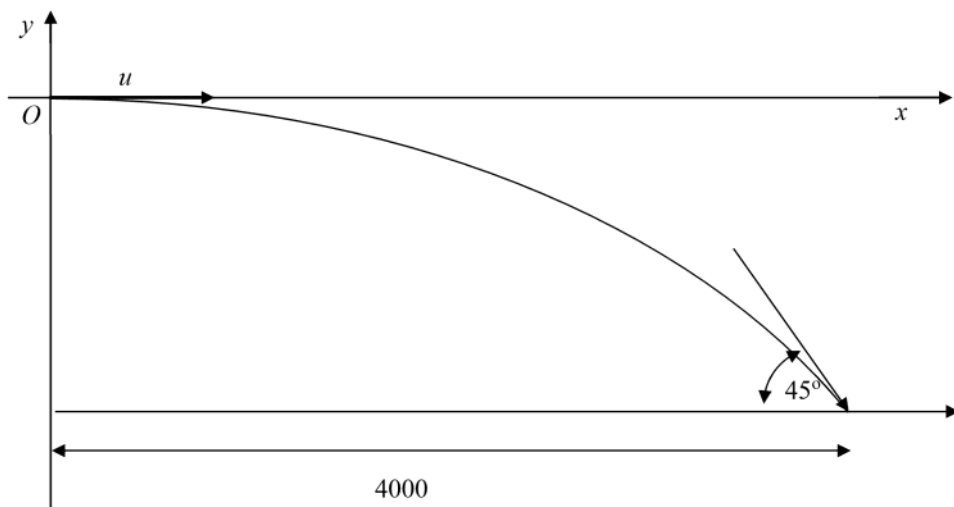
$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}.$$

- (b) Let $I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$ where n is a positive integer.

- (i) Prove that $I_{2n+1} = \frac{e}{2} - nI_{2n-1}$. 3

- (ii) Hence or otherwise, prove that $2 \int_0^1 x^{2n-1} (1+x^2) e^{x^2} dx \leq e$ for $n \geq 1$. 2

- (c) An aircraft flying horizontally at $u \text{ ms}^{-1}$ delivers an emergency medical supply package that hits the ground 4000 metres away, measured horizontally. The package experiences an air resistance of $0.1v$ where v is the velocity at time t and g is the acceleration due to gravity. The package hits the ground at an angle of 45° to the horizontal.



You can assume that after t seconds after release the position vector is given by

$$\mathbf{r}(t) = \begin{pmatrix} 10u(1 - e^{-0.1t}) \\ 100g(1 - e^{-0.1t}) - 10gt \end{pmatrix}. \quad (\text{Do not prove this result})$$

- (i) Show that the velocity vector $\mathbf{v}(t)$ of the particle is given by

1

$$\mathbf{v}(t) = \begin{pmatrix} ue^{-0.1t} \\ -10g(1 - e^{-0.1t}) \end{pmatrix}.$$

- (ii) Find the time when the package hits the ground and the speed on impact, where
 $g = 10 \text{ ms}^{-2}$. **2**

End of Exam

$$(i) \quad I_1 = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \quad I_2 = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

let $u = \pi - x$ $x = \pi, u = 0$

$\frac{du}{dx} = -1$ $x = 0, u = \pi$

$$I_1 = \int_{\pi}^0 \frac{(\pi - u) \sin(\pi - u)}{1 + [\cos(\pi - u)]^2} (-du) \quad (1)$$

flips limits.

$$\left. \begin{aligned} \sin(\pi - \theta) &= \sin \theta \\ \cos(\pi - \theta) &= -\cos \theta \end{aligned} \right\} \begin{aligned} &\text{(This is an angle} \\ &\text{in the second quadrant)} \end{aligned}$$

$$= \int_0^{\pi} \frac{(\pi - u) \sin u}{1 + (-\cos u)^2} du$$

$$= \int_0^{\pi} \frac{(\pi - u) \sin u}{1 + \cos^2 u} du$$

As this is a definite integral, the variable of integration can be changed.

$$= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx \quad \text{As required} \quad (1)$$

(ii)

$$I_1 = I_2$$

$$I_2 = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} - \frac{x \sin x}{1 + \cos^2 x} dx$$

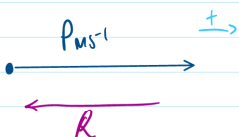
$$\therefore I_1 = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - I_1$$

$$2I_1 = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx \quad (1)$$

$$\begin{aligned} \text{let } u &= \cos x & x=\pi, u &= -1 \\ \frac{du}{dx} &= -\sin x & x=0, u &= 1 \\ \therefore -du &= \sin x \, dx \end{aligned}$$

$$\begin{aligned} &= -\frac{\pi}{2} \int \frac{1}{1+u^2} du \\ &= -\frac{\pi}{2} \left[\tan^{-1}(u) \right]_{-1}^1 + C \\ &= -\frac{\pi}{2} \left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \right] \\ &= -\frac{\pi^2}{4} \end{aligned}$$

(1)

(b) (i) 

let $u = P$
 $R = v + v^3$

$\Sigma F = -R$
 $m \ddot{x} = -(v + v^3) \quad m=1$

(1)

$$\therefore \ddot{x} = -v(1+v^2)$$

(ii) $v \cdot \frac{dv}{dx} = \ddot{x}$

$$\therefore v \cdot \frac{dv}{dx} = -v(1+v^2)$$

$$\frac{-dv}{1+v^2} = dx$$

$$\int_p^v \frac{-dv}{1+v^2} = \int_0^x dx$$

$$\begin{aligned} \therefore x &= -\left[\tan^{-1}(v) \right]_p^v \\ &= -\tan^{-1}(v) + \tan^{-1}(p) \end{aligned}$$

(1)

$$= \tan^{-1}\left(\frac{p-v}{1+pv}\right) \quad \text{As required.}$$

$$\tan \alpha = A \quad \tan \beta = B$$

$$\therefore \alpha + \beta = \tan^{-1} A - \tan^{-1} B$$

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ &= \frac{A - B}{1 + AB} \end{aligned}$$

$$\therefore \alpha + \beta = \tan^{-1}\left(\frac{A-B}{1+AB}\right)$$

(1)

(iii) $\ddot{x} = \frac{dv}{dt}$

$$\therefore \frac{dr}{dt} = -v(1+v^2)$$

$$\therefore dt = \frac{-dr}{v(1+v^2)}$$

$$\int_0^t dt = - \int_p^r \frac{1}{v(1+v^2)} dr \quad (1)$$

$$\text{let } \frac{A}{v} + \frac{Bv+C}{1+v^2} = \frac{1}{v(1+v^2)}$$

$$\therefore 1 = A(1+v^2) + v(Bv+C)$$

$$\text{When } v=0 \quad 1 = A$$

$$v=1 \quad 1 = 2A + B + C \quad \therefore B+C = -1$$

$$v=-1 \quad 1 = 2A + B - C \quad B-C = -1$$

$$\therefore C=0 \\ B=-1$$

$$t = - \int_p^r \frac{1}{v} - \frac{v}{1+v^2} dr \quad (1)$$

$$= - \left[\ln v - \frac{1}{2} \ln |1+v^2| \right]_p^r$$

$$= \frac{1}{2} \left[\ln |1+v^2| - \ln v^2 \right]_p^r$$

$$= \frac{1}{2} \left[\ln \left| \frac{1+v^2}{v^2} \right| \right]_p^r$$

$$= \frac{1}{2} \left[\ln \left| \frac{1+v^2}{v^2} \right| - \ln \left| \frac{1+p^2}{p^2} \right| \right]$$

$$t = \frac{1}{2} \ln \left| \frac{p^2(1+v^2)}{v^2(1+p^2)} \right|$$

$$\therefore \frac{p^2(1+v^2)}{v^2(1+p^2)} = e^{2t}$$

$$p^2(1+v^2) = v^2(1+p^2)e^{2t}$$

$$p^2 + p^2v^2 = v^2(1+p^2)e^{2t}$$

$$p^2 = v^2(1+p^2)e^{2t} - p^2v^2$$

$$= v^2([1+p^2]e^{2t} - p^2)$$

$$v^2 = \frac{p^2}{(1+p^2)e^{2t} - p^2}$$

As required.

$$(c) \text{ i) } x = 3 + \sin^2 t \quad \therefore \sin^2 t = x-3$$

$$\frac{dx}{dt} = 2 \cos t \sin t$$

$$\frac{d^2x}{dt^2} = 2 \cos t \times \cos t + \sin t \times -2 \sin t \\ = 2 \cos^2 t - 2 \sin^2 t$$

a''

$$\begin{aligned} &= 2\cos^2 t - 2\sin^2 t \\ &= 2(1 - \sin^2 t) - 2\sin^2 t \\ &= 2 - 4\sin^2 t \\ &= 2 - 4(x-3) \\ &= 2 - 4x + 12 \\ &= 14 - 4x \\ &= 4\left[\frac{7}{2} - x\right] \end{aligned}$$

①

This SHM as the acceleration is proportional to, but in the opposite direction of, the displacement from $\frac{7}{2}$. ①

(ii) $n=2$ $\therefore T = \frac{2\pi}{2} = \pi$

(iii) π seconds is one period $\therefore$ It will travel four times the amplitude.

If $x = 3 + \sin^2 t$ then max value for $x = 4$ & min value is 3

$\therefore$ Amplitude is $\frac{1}{2}$ cm

$\therefore$ Travels 2 cm ①

① Justifying the answer.

(a) Sum of $\hat{i} - \lambda \hat{k}$ & $\lambda \hat{i} + 2\hat{j} - \hat{k} = (\lambda+1)\hat{i} + 2\hat{j} + (-1-\lambda)\hat{k}$

$$\begin{pmatrix} 1 \\ -2\lambda \\ -1 \end{pmatrix} \cdot \begin{pmatrix} \lambda+1 \\ 2 \\ -1-\lambda \end{pmatrix} = \lambda+1 - 4\lambda + 1 + \lambda$$

$$\textcircled{1} \quad 6 = -2\lambda + 2$$

$$-2\lambda = 4$$

$$\lambda = -2 \quad \textcircled{1}$$

(b) (i) If $\vec{AC}$ perpendicular to $\vec{BD}$ then $\vec{AC} \cdot \vec{BD} = 0$

$$\vec{AC} = \begin{pmatrix} 4 - (-4) \\ -1 - 3 \\ -1 - 3 \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix}$$

$$\vec{BD} = \begin{pmatrix} 0 - 4 \\ 2 - \lambda \\ -6 - 6 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 - \lambda \\ -12 \end{pmatrix}$$

$$\begin{aligned} \vec{AC} \cdot \vec{BD} &= 8 \times -4 + (-4)(2 - \lambda) + (-4)(-12) \\ &= -32 - 8 + 4\lambda + 48 \\ 0 &= 8 + 4\lambda \\ 4\lambda &= -8 \\ \lambda &= -2 \end{aligned} \quad \textcircled{1}$$

(ii) The line $\vec{AC} = \begin{pmatrix} -4 \\ 3 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix}$

The line $\vec{BD} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} + \lambda_2 \begin{pmatrix} -4 \\ 4 \\ -12 \end{pmatrix}$ ①

Solve simultaneously $x: -4 + 8\lambda_1 = 4 - 4\lambda_2$

$$-1 + 2\lambda_1 = 1 - \lambda_2$$

$$2\lambda_1 + \lambda_2 = 2$$

$y: 3 - 4\lambda_1 = -2 + 4\lambda_2$

$$5 = 4\lambda_1 + 4\lambda_2$$

Solve

$$2\lambda_1 + \lambda_2 = 2 \quad 4\lambda_1 + 4\lambda_2 = 5 \quad \textcircled{2}$$

$$8\lambda_1 + 4\lambda_2 = 8 \quad \textcircled{1}$$

$$\textcircled{1} - \textcircled{2}$$

$$4\lambda_1 = 3$$

$$\lambda_1 = \frac{3}{4} \quad \therefore \lambda_2 = \frac{1}{2} \quad \textcircled{1}$$

Check with $z: 3 - 4\lambda_1 = 6 - 12\lambda_2$

$$3 - 3 = 6 - 6$$

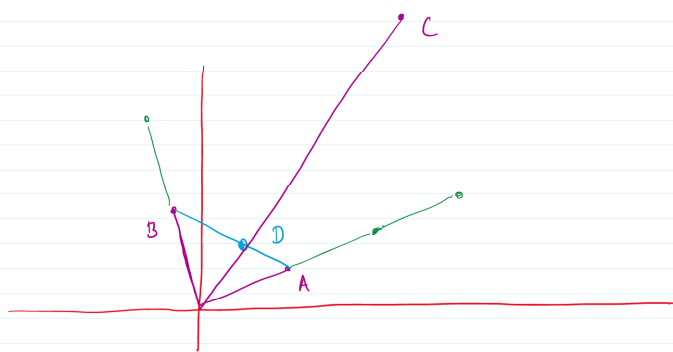
$\therefore \lambda_1 = \frac{3}{4}$ & $\lambda_2 = \frac{1}{2}$ at the point of intersection.

$$\therefore \vec{r} = \begin{pmatrix} -4 + 8\lambda_1 \\ 3 - 4\lambda_1 \\ 3 - 4\lambda_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \quad \therefore \vec{r} = 2\hat{i} \quad \textcircled{1}$$

(c) (i) $\vec{OA} = \underline{a}$ $\vec{OB} = \underline{b}$ $\vec{OC} = 3\underline{a} + 2\underline{b}$

/ c

(c) (i) $\vec{OA} = \underline{a}$ $\vec{OB} = \underline{b}$ $\vec{OC} = 3\underline{a} + 2\underline{b}$



$$\vec{OD} = \frac{1}{5}(3\underline{a} + 2\underline{b})$$

$$\vec{AB} = \underline{b} - \underline{a} \quad (1)$$

$$\vec{AD} = \frac{1}{5}(3\underline{a} + 2\underline{b}) - \underline{a}$$

$$= -\frac{2}{5}\underline{a} + \frac{2}{5}\underline{b}$$

$$= \frac{2}{5}(\underline{b} - \underline{a}) \quad (1)$$

$$= \frac{2}{5}\vec{AB} \quad \therefore D \text{ lies on } \vec{AB} \text{ as}$$

$\vec{AD}$ is parallel to $\vec{AB}$ but shorter (multiplied by a constant that is $0 < k < 1$) (1)

(ii) $\vec{AD} = \frac{2}{5}\vec{AB}$ which means $\vec{DB} = \frac{3}{5}\vec{AB}$ (1)

$\therefore D$ is closer to A as $|\vec{AD}| < |\vec{DB}|$

(d) (i) $z = \cos\theta + i\sin\theta$

$$z^n = \cos n\theta + i\sin n\theta \quad (\text{By DMT})$$

$$z^{-n} = \cos(-n\theta) + i\sin(-n\theta) \quad \text{But, } \cos\theta \text{ is an even function \& \sin\theta \text{ is an odd function}$$

$$z^{-n} = \cos(n\theta) - i\sin(n\theta)$$

$$\therefore z^n + z^{-n} = \cancel{\cos(n\theta)} + i\sin(n\theta) - [\cancel{\cos(n\theta)} - i\sin(n\theta)]$$

$$= 2i\sin(n\theta) \text{ As required.} \quad (1)$$

(ii) $(z - \frac{1}{z})^5 = z^5 - 5 \times z^4 \times \frac{1}{z} + 10 \times z^3 \times \frac{1}{z^2} - 10 \times z^2 \times \frac{1}{z^3} + 5 \times z \times \frac{1}{z^4} - \frac{1}{z^5}$

$$= (z^5 - \frac{1}{z^5}) - 5 \left[z^3 - \frac{1}{z^3} \right] + 10 \left[z - \frac{1}{z} \right] \quad (1)$$

$$= 2i\sin 5\theta - 5 \left[2i\sin 3\theta \right] + 10 \left[2i\sin \theta \right]$$

$$= 2i \sin 5\theta - 5 \left[2i \sin 3\theta \right] + 10 \left[2i \sin \theta \right]$$

$$= 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta \quad (1)$$

$$\left(2 - \frac{1}{2}\right)^5 = \left(2i \sin \theta\right)^5$$

$$= 32i \sin^5 \theta$$

$$\therefore 32i \sin^5 \theta = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$$

$$\sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta \quad \text{As required.}$$

$$(a) (i) (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) = |\underline{a}|^2 + |\underline{b}|^2$$

$$\text{LHS} = \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b} \quad (1)$$

$$\text{If } \underline{a} \perp \underline{b} \text{ then } \underline{a} \cdot \underline{b} = 0 \quad (1)$$

$$= \underline{a} \cdot \underline{a} + \underline{b} \cdot \underline{b}$$

$$\underline{a} \cdot \underline{a} = |\underline{a}|^2$$

$$= |\underline{a}|^2 + |\underline{b}|^2 \quad \text{As required.}$$

(ii)



$$\vec{LM} = \underline{a} \quad \vec{MN} = \underline{b} \quad \underline{a} \text{ and } \underline{b} \text{ are perpendicular.}$$

$$\therefore \vec{LN} = \underline{a} + \underline{b}$$

$$|\vec{LN}|^2 = \vec{LN} \cdot \vec{LN} \quad (1)$$

$$= (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b})$$

$$= |\underline{a}|^2 + |\underline{b}|^2 \quad (\text{see part (i)}) \quad (1)$$

$$= |\vec{LM}|^2 + |\vec{MN}|^2 \quad \text{As required.}$$

$$(b) \int x^2 \sqrt{1-x^2} \, dx$$

$$\text{let } x = \sin \theta$$

$$\frac{dx}{d\theta} = \cos \theta$$

$$\therefore I = \int \sin^2 \theta \sqrt{1 - \sin^2 \theta} \, d\theta \cdot \cos \theta \quad (1)$$

$$= \int \sin^2 \theta \cos^2 \theta \, d\theta$$

$$= \int (\sin \theta \cos \theta)^2 \, d\theta$$

$$= \int \left(\frac{\sin 2\theta}{2} \right)^2 \, d\theta$$

$$= \frac{1}{4} \int \sin^2 2\theta \, d\theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$= 1 - 2\sin^2 \theta$$

$$= \frac{1}{8} \int 1 - \cos 4\theta \, d\theta \quad \therefore \sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta) \quad (1)$$

$$= \frac{1}{8} \left[\theta - \frac{\sin 4\theta}{4} \right] + C$$

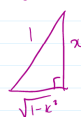
$$= \frac{1}{8} \left[\sin^{-1}(x) - 4(x\sqrt{1-x^2})(1-x^2) \right]$$

$$= \frac{1}{8} \left[\sin^{-1}(x) - 4x\sqrt{1-x^2}(1-2x^2) \right] + C$$

$$\sin 4\theta = 2\sin 2\theta \cos 2\theta$$

$$= 2[2\sin \theta \cos \theta][\cos^2 \theta - \sin^2 \theta]$$

$$\text{if } \sin \theta = x \quad \cos \theta = \sqrt{1-x^2}$$

(1) In terms of x

(c) (i)



$$R = kv^2$$

$$\Sigma F = mg - R$$

$$\therefore m\ddot{x} = mg - R \quad \text{But } m=1$$

$$\therefore \ddot{x} = g - R$$

$$v \cdot \frac{dv}{dx} = g - kv^2$$

$$\therefore dx = \frac{v}{g - kv^2} dv \quad (1)$$

$$\int_0^x dx = \int_0^v \frac{v}{g - kv^2} dv$$

$$x = -\frac{1}{2k} \int_0^v \frac{2kv}{g - kv^2} dv$$

$$= -\frac{1}{2k} \left[\ln |g - kv^2| \right]_0^v$$

$$= -\frac{1}{2k} \left[\ln |g - kv^2| - \ln |g| \right]$$

$$= \frac{1}{2k} \ln \left| \frac{g}{g - kv^2} \right| \quad \text{As required.}$$

$$(ii) \quad \frac{dv}{dt} = g - kv^2$$

$$\therefore dt = \frac{1}{g - kv^2} dv$$

$$\int_0^t dt = \int_0^v \frac{1}{2\sqrt{g}} \left(\frac{1}{\sqrt{g} + \sqrt{k}v} + \frac{1}{\sqrt{g} - \sqrt{k}v} \right) dv$$

$$t = \frac{1}{2\sqrt{kg}} \left[\ln \left| \frac{\sqrt{g} + \sqrt{k}v}{\sqrt{g} - \sqrt{k}v} \right| \right]_0^v$$

$$= \frac{1}{2\sqrt{kg}} \ln \left| \frac{\sqrt{g} + \sqrt{k}v}{\sqrt{g} - \sqrt{k}v} \right| \quad (1)$$

$$\text{let } \frac{1}{(\sqrt{g} - \sqrt{k}v)(\sqrt{g} + \sqrt{k}v)} = \frac{A}{\sqrt{g} + \sqrt{k}v} + \frac{B}{\sqrt{g} - \sqrt{k}v}$$

$$1 = A(\sqrt{g} - \sqrt{k}v) + B(\sqrt{g} + \sqrt{k}v)$$

$$\text{When } v=0 \quad 1 = \sqrt{g}(A+B) \quad \dots (1)$$

$$\text{When } v=1 \quad 1 = \sqrt{g}(A+B) + \sqrt{k}(B-A) \quad \dots (2)$$

$$(2) - (1)$$

$$\sqrt{k}(B-A) = 0$$

$$\therefore B=A$$

$$\therefore 2\sqrt{g}A = 1$$

$$A = \frac{1}{2\sqrt{g}}$$

$$\therefore \frac{1}{g - kv^2} = \frac{1}{2\sqrt{g}} \left(\frac{1}{\sqrt{g} + \sqrt{k}v} + \frac{1}{\sqrt{g} - \sqrt{k}v} \right)$$

$$\text{Find } t \text{ when } v = \frac{U}{2} \quad (\text{let } U = p)$$

$$\therefore t = \frac{1}{2\sqrt{kg}} \ln \left| \frac{\sqrt{g} + \sqrt{k} \cdot \frac{p}{2}}{\sqrt{g} - \sqrt{k} \cdot \frac{p}{2}} \right| \quad (1)$$

However, As p is terminal velocity

$$kp^2 = g \quad (\text{As terminal velocity occurs when } \Sigma F = 0)$$

$$p^2 = \frac{g}{k}$$

$$p = \sqrt{\frac{g}{k}}$$

$$t = \frac{1}{2\sqrt{kg}} \ln \left[\frac{\sqrt{g} + \sqrt{k} \cdot \frac{p}{2}}{\sqrt{g} - \sqrt{k} \cdot \frac{p}{2}} \right] \quad \div \sqrt{k}$$

$$= \frac{1}{2\sqrt{kg}} \ln \left[\frac{\sqrt{\frac{g}{k}} + \frac{p}{2}}{\sqrt{\frac{g}{k}} - \frac{p}{2}} \right]$$

$$= \frac{1}{2\sqrt{kg}} \ln \left[\frac{p + \frac{p}{2}}{p - \frac{p}{2}} \right]$$

$$= \frac{1}{2\sqrt{kg}} \ln \left| \frac{\frac{3p}{2}}{\frac{p}{2}} \right|$$

$$= \frac{1}{2\sqrt{kg}} \ln 3$$

$$= \frac{1}{2} \times \sqrt{\frac{g}{k}} \times \frac{1}{g} \ln 3$$

$$= \frac{p}{2g} \ln 3 \quad \text{As required.}$$

①

$$(d) \int (1+2x^2)e^{x^2} dx = \int e^{x^2} dx + \int 2x^2 e^{x^2} dx$$

$$I = \int e^{x^2} dx \quad J = \int 2x^2 e^{x^2} dx \quad \text{①}$$

$$u = x$$

$$\frac{du}{dx} = 1$$

$$v = e^{x^2}$$

$$\frac{dv}{dx} = 2xe^{x^2}$$

$$J = xe^{x^2} - \int e^{x^2} dx \quad \text{①}$$

$$= xe^{x^2} - I$$

$$\therefore J + I = xe^{x^2} + C$$

①

(a) (i) $z^7 = 1$ let $z = e^{i\theta}$ As $|z|=1$

$$e^{i7\theta} = e^0, e^{\pm 2\pi i}, e^{\pm 4\pi i}, e^{\pm 6\pi i}$$

$$\therefore e^{i\theta} = e^0, e^{\pm \frac{2\pi}{7}i}, e^{\pm \frac{4\pi}{7}i}, e^{\pm \frac{6\pi}{7}i}$$

① Answers
① Exp. form

(ii) let $z = \cos\theta + i\sin\theta$ ($|z|=1$)

$$z^n = \cos n\theta + i\sin n\theta$$

$$z^{-n} = \cos(-n\theta) + i\sin(-n\theta) \quad \left. \vphantom{\begin{matrix} z^n \\ z^{-n} \end{matrix}} \right\} \text{By DMT}$$

But, $\cos$ is an even function & $\sin$ is an odd function

$$\therefore z^{-n} = \cos n\theta - i\sin n\theta$$

$$\begin{aligned} z^n + z^{-n} &= \cancel{\cos n\theta} + \cancel{i\sin n\theta} + \cos n\theta - \cancel{i\sin n\theta} \\ &= 2\cos n\theta \end{aligned}$$

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0 \quad \text{has roots } e^{\pm \frac{2\pi}{7}i}, e^{\pm \frac{4\pi}{7}i}, e^{\pm \frac{6\pi}{7}i}$$

$$\div z^3 \quad (z \neq 0)$$

$$z^3 + z^2 + z + 1 + z^{-1} + z^{-2} + z^{-3} = 0 \quad \text{①}$$

$$\text{LHS} = z^3 + z^{-3} + z^2 + z^{-2} + z + z^{-1} + 1$$

Using ①

$$= (2\cos 3\theta) + (2\cos 2\theta) + (2\cos \theta) + 1 = 0 \quad \text{As required} \quad \text{①}$$

(iii) $2\cos 3\theta + 2\cos 2\theta + 2\cos \theta = -1$

$$\cos 3\theta + \cos 2\theta + \cos \theta = -\frac{1}{2}$$

$$\text{Solutions are } \theta = \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}$$

$$\text{let } \theta = \frac{2\pi}{7}$$

$$\cos 3\left(\frac{2\pi}{7}\right) + \cos 2\left(\frac{2\pi}{7}\right) + \cos \left(\frac{2\pi}{7}\right) = -\frac{1}{2}$$

$$\left(\text{As } \theta = \frac{2\pi}{7} \text{ is a solution} \right)$$

$$\cos \frac{6\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} = -\frac{1}{2} \quad \text{As required.}$$

① substitution

① Rearrange

(b) (i) $I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$

$$(b) \quad (i) \quad I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$$

$$= \int_0^1 x^{2n} \cdot x e^{x^2} dx$$

$$u = x^{2n}$$

$$v = \frac{e^{x^2}}{2}$$

$$\frac{du}{dx} = 2nx^{2n-1}$$

$$\frac{dv}{dx} = x e^{x^2}$$

$$\therefore I_{2n+1} = \left[\frac{x^{2n} e^{x^2}}{2} \right]_0^1 - \int_0^1 \frac{2n x^{2n-1} e^{x^2}}{2} dx$$

$$= \left[\frac{1 \times e}{2} - \frac{0 \times 1}{2} \right] - n \int_0^1 x^{2n-1} e^{x^2} dx$$

$$= \frac{e}{2} - n I_{2n-1} \quad \text{As required.}$$

$$(ii) \quad I_{2n+1} = \frac{e}{2} - n I_{2n-1}$$

$$\therefore I_{2n+1} + n I_{2n-1} = \frac{e}{2}$$

As $n \geq 1$

$$\left. \begin{aligned} I_{2n+1} + I_{2n-1} &\leq I_{2n+1} + n I_{2n-1} \\ &\leq \frac{e}{2} \end{aligned} \right\} \textcircled{1}$$

$$\text{LHS} = \left[\int_0^1 x^{2n+1} e^{x^2} dx + \int_0^1 x^{2n-1} e^{x^2} dx \right] \textcircled{1}$$

$$= \int_0^1 x^{2n+1} e^{x^2} + x^{2n-1} e^{x^2} dx$$

$$= \int_0^1 x^{2n-1} [x^2 e^{x^2} + e^{x^2}] dx$$

$$= \int_0^1 x^{2n-1} e^{x^2} (1 + x^2) dx$$

$$\therefore \int_0^1 x^{2n-1} e^{x^2} (1 + x^2) dx \leq \frac{e}{2}$$

$$\therefore \int_0^1 x^{2n-1} e^{x^2} (1+x^2) dx \leq \frac{e}{2}$$

$$2 \int_0^1 x^{2n-1} e^{x^2} (1+x^2) dx \leq e \quad \text{is required.}$$

(c) (i) $s(t) = \begin{cases} 10u(1-e^{-0.1t}) \\ 100g(1-e^{-0.1t}) - 10gt \end{cases}$

$$\frac{d}{dt}(s(t)) = v(t)$$

$$\begin{aligned} \frac{d}{dt} 10u(1-e^{-0.1t}) &= 10u - 10ue^{-0.1t} \\ &= 0 - 10u \times -0.1 e^{-0.1t} \\ &= ue^{0.1t} \end{aligned}$$

$$\begin{aligned} \frac{d}{dt} 100g(1-e^{-0.1t}) - 10gt &= \frac{d}{dt} (100g - 100ge^{-0.1t} - 10gt) \\ &= 0 - 100g \times -0.1 e^{-0.1t} - 10g \\ &= 10ge^{-0.1t} - 10g \\ &= -10g [1 - e^{-0.1t}] \end{aligned}$$

(1)

(ii) If it hits the ground at 45° then
Horizontal velocity = Vertical velocity

$$\begin{aligned} ue^{-0.1t} &= 100(1-e^{-0.1t}) \quad (\text{NB, changed to positive as it is travelling down.}) \\ ue^{-0.1t} &= 100 - 100e^{-0.1t} \\ \therefore (u+100)e^{-0.1t} &= 100 \\ e^{-0.1t} &= \frac{100}{u+100} \quad \dots (1) \end{aligned}$$

When it hits the ground, it has travelled 4000m horizontally. $\therefore x(t) = 4000$

$$4000 = 10u(1-e^{-0.1t}) \quad \dots (2)$$

(1) $\Rightarrow$ (2)

$$4000 = 10u \left[1 - \frac{100}{u+100} \right]$$

$$= 10u \left[\frac{u+100 - 100}{u+100} \right]$$

$$400u + 40000 = u^2$$

$$u^2 - 400u + 40000 = 0$$

$$u = \frac{400 \pm \sqrt{(400)^2 + 4 \times 1 \times 40000}}{2}$$

$$= \frac{400 \pm 400\sqrt{2}}{2}$$

$$= 200 \pm 200\sqrt{2} \quad (\text{As } u > 0)$$

$$u = 200 + 200\sqrt{2}$$

$$t = -10 \ln \left| \frac{100}{100 + u} \right|$$

$$\doteq 17.6 \text{ seconds} \quad (1 \text{ dp}) \quad (1)$$

let speed of input = v

$$|v|^2 = |\dot{x}|^2 + |\dot{y}|^2 \quad \text{But, } |\dot{x}| = |\dot{y}|$$

$$\therefore |\underline{v}|^2 = 2|\dot{x}|^2$$

$$v = \sqrt{2} \left[(200 + 200\sqrt{2}) e^{-0.1t} \right]$$

$$\doteq 117.16 \text{ ms}^{-1} \quad (2 \text{ dp}) \quad (1)$$